

memory pool. A memory pool consists of transactions that have been validated but not included in any of the blocks yet. When the new block needs to be added to the existing network, the Tendermint node takes care of the Byzantine fault-tolerant property.

Step 4: Writing transactions to MongoDB: Finally, the Tendermint node delivers the transaction to a BigchainDB to check its validity. Once the transaction is committed successfully, it is written to a MongoDB database.

Steps for installing the BigchainDB server

The BigchainDB server, MongoDB and Tendermint, must run in every BigchainDB node. The steps for installing the BigchainDB server, MongoDB and Tendermint on Ubuntu 18.04 Linux OS are described below.

Step 1: Installing the BigchainDB server: The BigchainDB server requires the Python 3.6+ version; so first install the required Python package, as follows:

```
# Installation of python latest
version on Ubuntu 18.04
```

```
$ sudo apt install -y python3-pip
libssl-dev
```

Now, the latest version of the BigchainDB server can be installed using the following command:

```
# 2.2.1 is the latest version of
BigchainDB server:
```

```
$ sudo pip3 install bigchaindb==2.2.1
```

The installed version of BigchainDB can be checked using the command given below:

```
$ bigchaindb -v
```

Step 2: Installing and starting

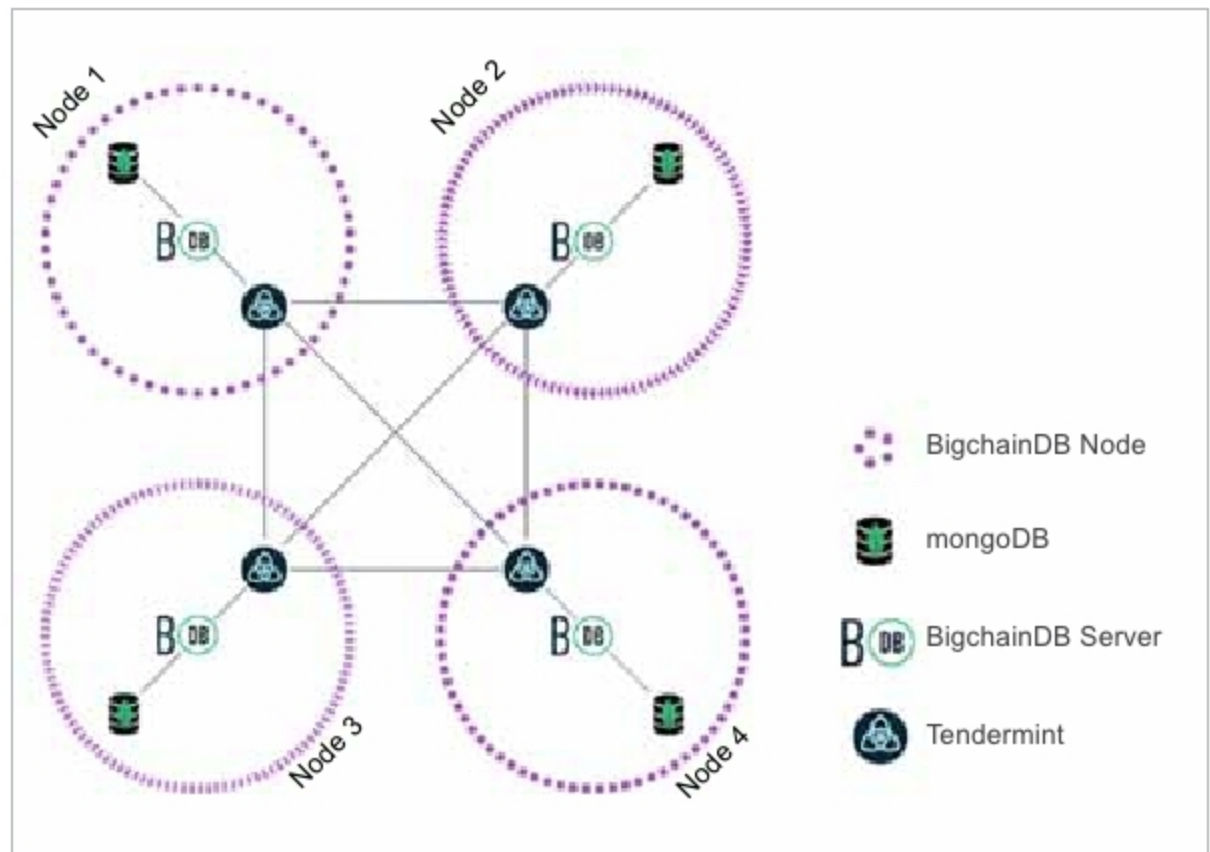


Figure 1: The four main components in the BigchainDB network

MongoDB: BigchainDB requires version 3.4 or higher for the MongoDB database; so ensure that by using the following command:

```
$ sudo apt install -y mongodb
```

```
#Check the status of database working
or not
```

```
$ sudo systemctl status mongodb
```

Step 3: Installing Tendermint:

The BigchainDB server works well with Tendermint 0.31.5 (not a higher version number). To install and start Tendermint, use the following commands:

```
$ sudo apt install -y unzip
```

```
$ wget
https://github.com/tendermint/
tendermint/releases/download/v0.31.5/
tendermint_v0.31.5_linux_amd64.zip
```

```
$ unzip tendermint_v0.31.5_linux_amd64.
zip
$ rm tendermint_v0.31.5_linux_amd64.zip
$ sudo mv tendermint /usr/local/bin
```

```
# Start Tendermint node
$ tendermint init
```

BigchainDB combines an enterprise-grade distributed database (MongoDB) with a production-ready consensus engine (Tendermint) to provide the key benefits of both distributed databases and blockchain technology. **END** 🐧

References

- [1] <https://www.bigchaindb.com/>
- [2] <https://www.bigchaindb.com/whitepaper/>
- [3] <https://github.com/bigchaindb/bigchaindb>

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